



US0D1091824S

(12) **United States Design Patent** (10) **Patent No.:** **US D1,091,824 S**
Kruse et al. (45) **Date of Patent:** **** Sep. 2, 2025**

(54) **CATHODE SHIELD**

(71) Applicant: **GE Precision Healthcare LLC**,
Wauwatosa, WI (US)

(72) Inventors: **Kevin S. Kruse**, Muskego, WI (US);
Sergio Lemaitre, West Milwaukee, WI (US)

(73) Assignee: **GE PRECISION HEALTHCARE LLC**, Wauwatosa, WI (US)

(**) Term: **15 Years**

(21) Appl. No.: **29/873,757**

(22) Filed: **Apr. 6, 2023**

(51) **LOC (15) Cl.** **24-01**

(52) **U.S. Cl.**
USPC **D24/158**

(58) **Field of Classification Search**
USPC D24/158, 177, 185-186, 216, 231
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,860,219 B2 12/2010 Clark et al.
9,177,754 B2 11/2015 Hansen et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 308879504 * 10/2024
CN 308879509 * 10/2024
(Continued)

OTHER PUBLICATIONS

Kruse, K. et al., "Cathode Shield," U.S. Appl. No. 29/873,755, filed Apr. 6, 2023, 23 pages.

(Continued)

Primary Examiner — Justin M Jonaitis

Assistant Examiner — Samantha K Pollack

(74) *Attorney, Agent, or Firm* — McCoy Russell LLP

(57) **CLAIM**

The ornamental design for a cathode shield as shown and described.

DESCRIPTION

FIG. 1 is a front view of a cathode shield according to the claimed design, with a portion of an X-ray tube shown in environment.

FIG. 2 is a rear view of the cathode shield, with the portion of the X-ray tube shown in environment.

FIG. 3 is a right side view of the cathode shield, with the portion of the X-ray tube shown in environment.

FIG. 4 is a left side view of the cathode shield, with the portion of the X-ray tube shown in environment.

FIG. 5 is a top view of the cathode shield, with the portion of the X-ray tube shown in environment.

FIG. 6 is a bottom view of the cathode shield, with the portion of the X-ray tube shown in environment.

FIG. 7 is a bottom view of the cathode shield.

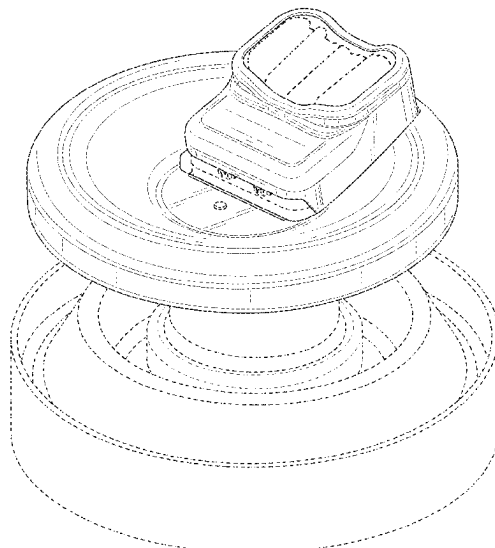
FIG. 8 is a top-front-left perspective view of the cathode shield, with a cathode and the portion of the X-ray tube shown in environment.

FIG. 9 is a sectional view of the cathode shield and the portion of the X-ray tube shown in environment, taken along line 9-9 in FIG. 5; and,

FIG. 10 is a sectional view of the cathode shield and the portion of the X-ray tube shown in environment, taken along line 10-10 in FIG. 5.

The dash-dash broken lines in the drawings showing the cathode and the portion of the X-ray tube illustrate environmental matter that forms no part of the claimed design. The dash-dot-dot sectional broken lines in FIG. 5 of the drawings form no part of the claimed design.

1 Claim, 10 Drawing Sheets



(58) **Field of Classification Search**

CPC H01J 1/025; H01J 1/52; H01J 1/92; H01J
1/94; H01J 35/06; H01J 35/14; H01J
35/16; H01J 2229/863; H01J 2229/887
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

D980,814	S	*	3/2023	Shin		D13/184
D981,973	S	*	3/2023	Kim		D13/184
D1,012,873	S	*	1/2024	Kim		D13/184
D1,058,525	S	*	1/2025	Kruse		D24/158
2002/0001068	A1	*	1/2002	Iwanczyk		A61B 6/4057 353/121
2015/0022080	A1	*	1/2015	Umbach		H01J 35/06 315/12.1

FOREIGN PATENT DOCUMENTS

EM	015036207-0001	*	12/2023
EM	015036207-0002	*	12/2023
EM	015036320-0001	*	12/2023

OTHER PUBLICATIONS

Kruse, K. et al., "Cathode Shield," U.S. Appl. No. 29/873,756, filed Apr. 6, 2023, 9 pages.

Kruse, K., "X-Ray Cathode Shield," U.S. Appl. No. 18/296,943, filed Apr. 6, 2023, 44 pages.

Kruse, K., "X-Ray Cathode Shield," U.S. Appl. No. 18/296,949, filed Apr. 6, 2023, 44 pages.

* cited by examiner

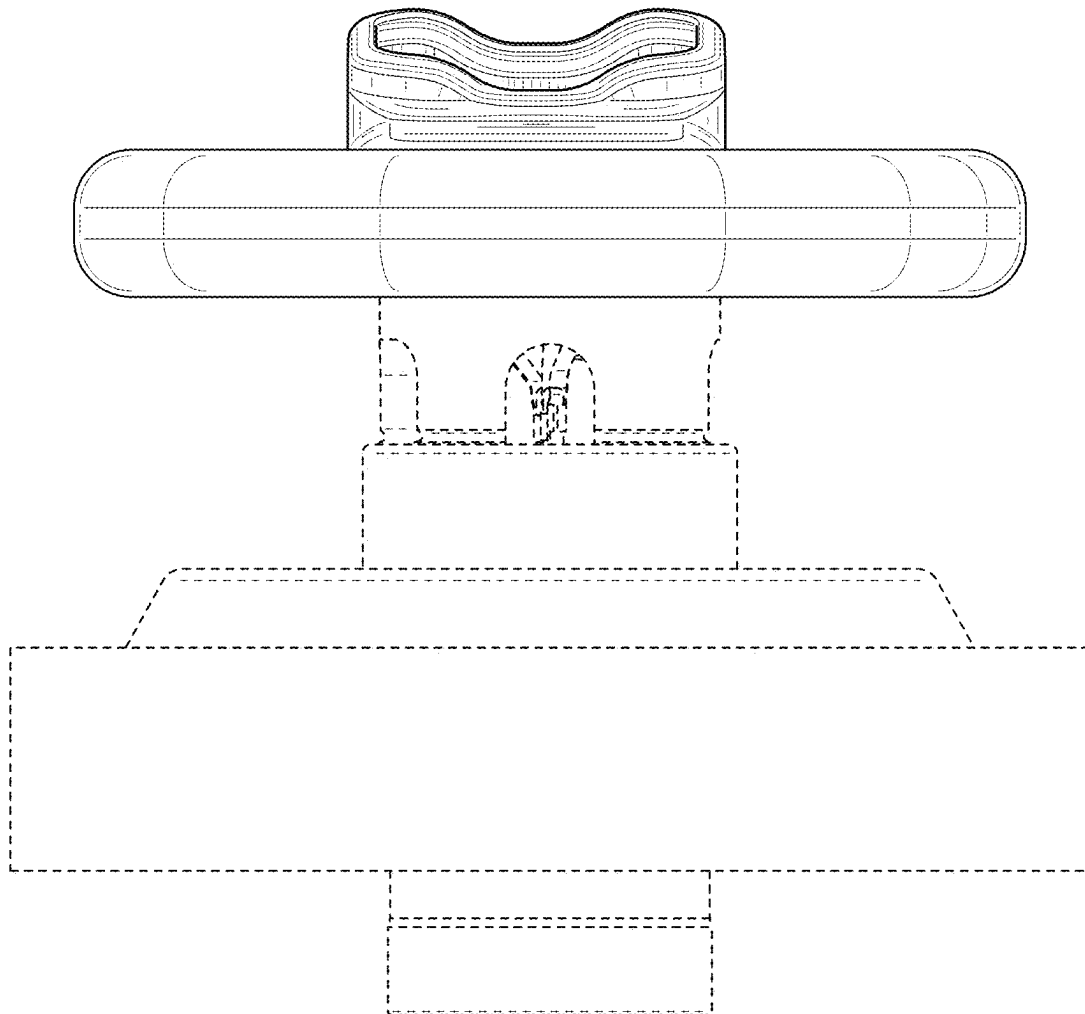


FIG. 1

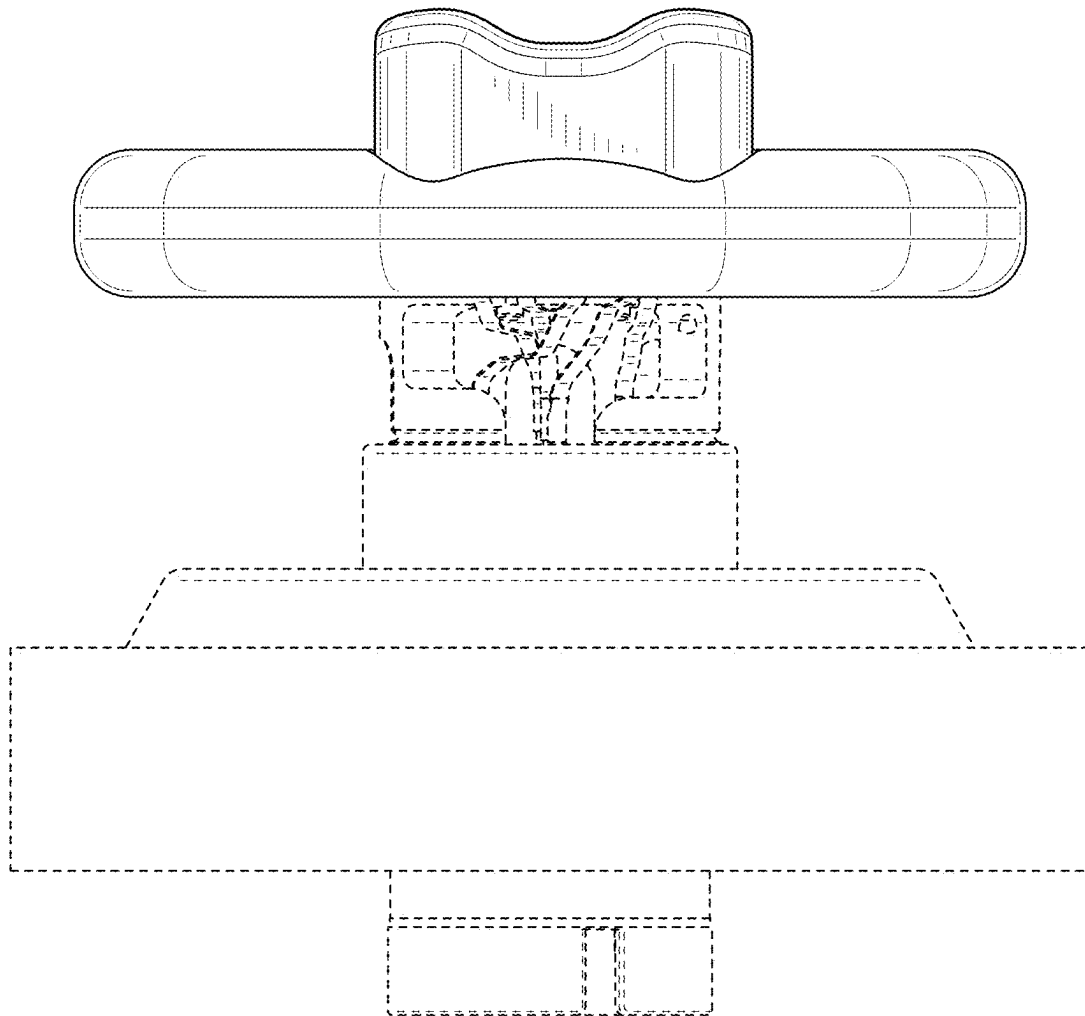


FIG. 2

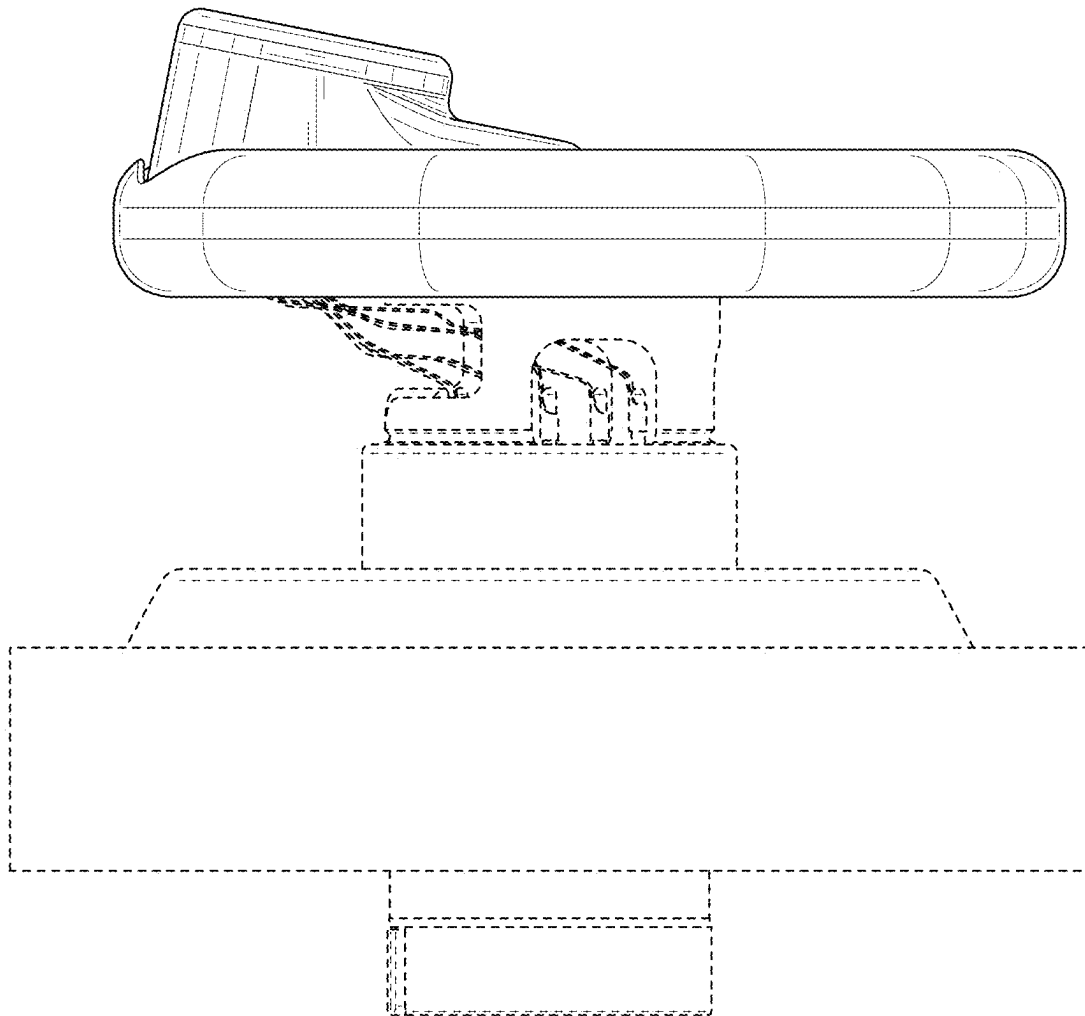


FIG. 3

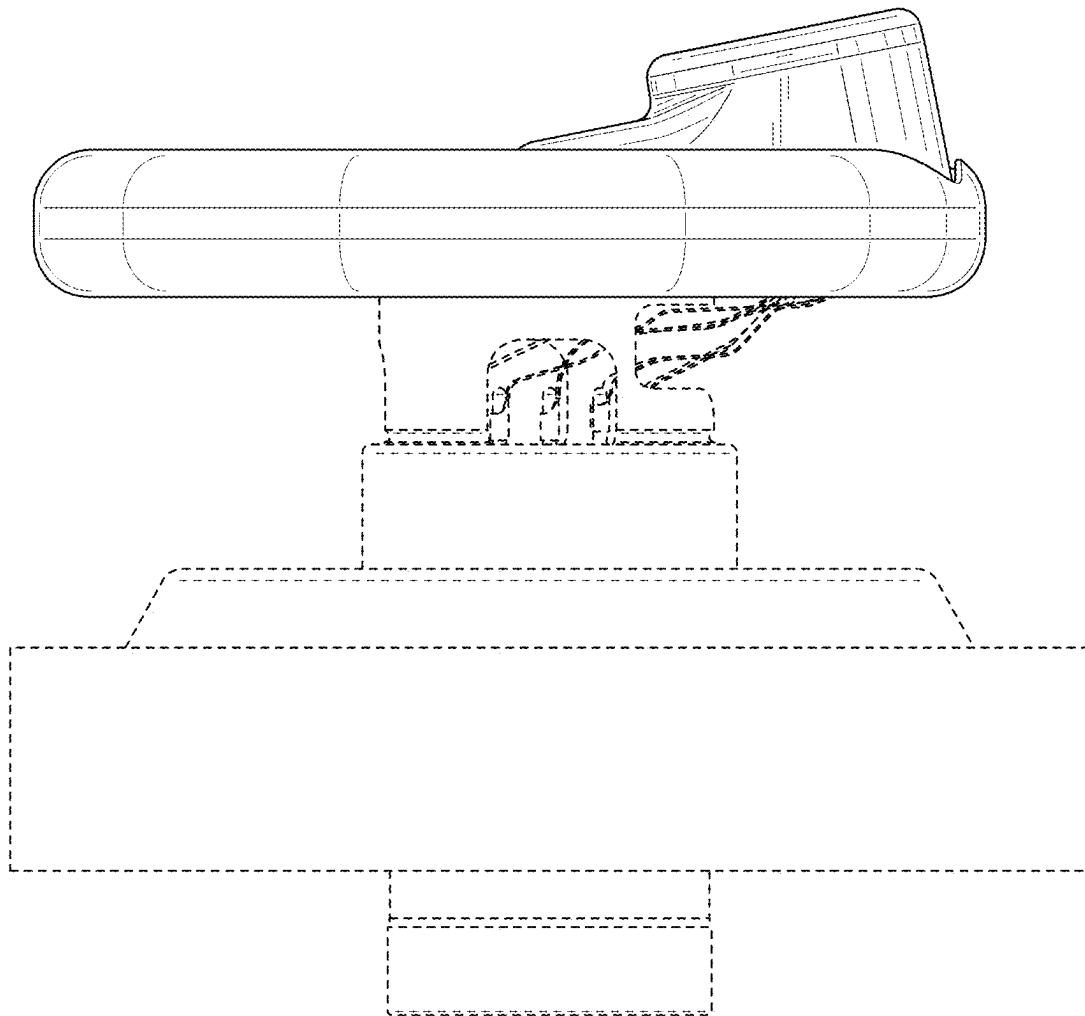


FIG. 4

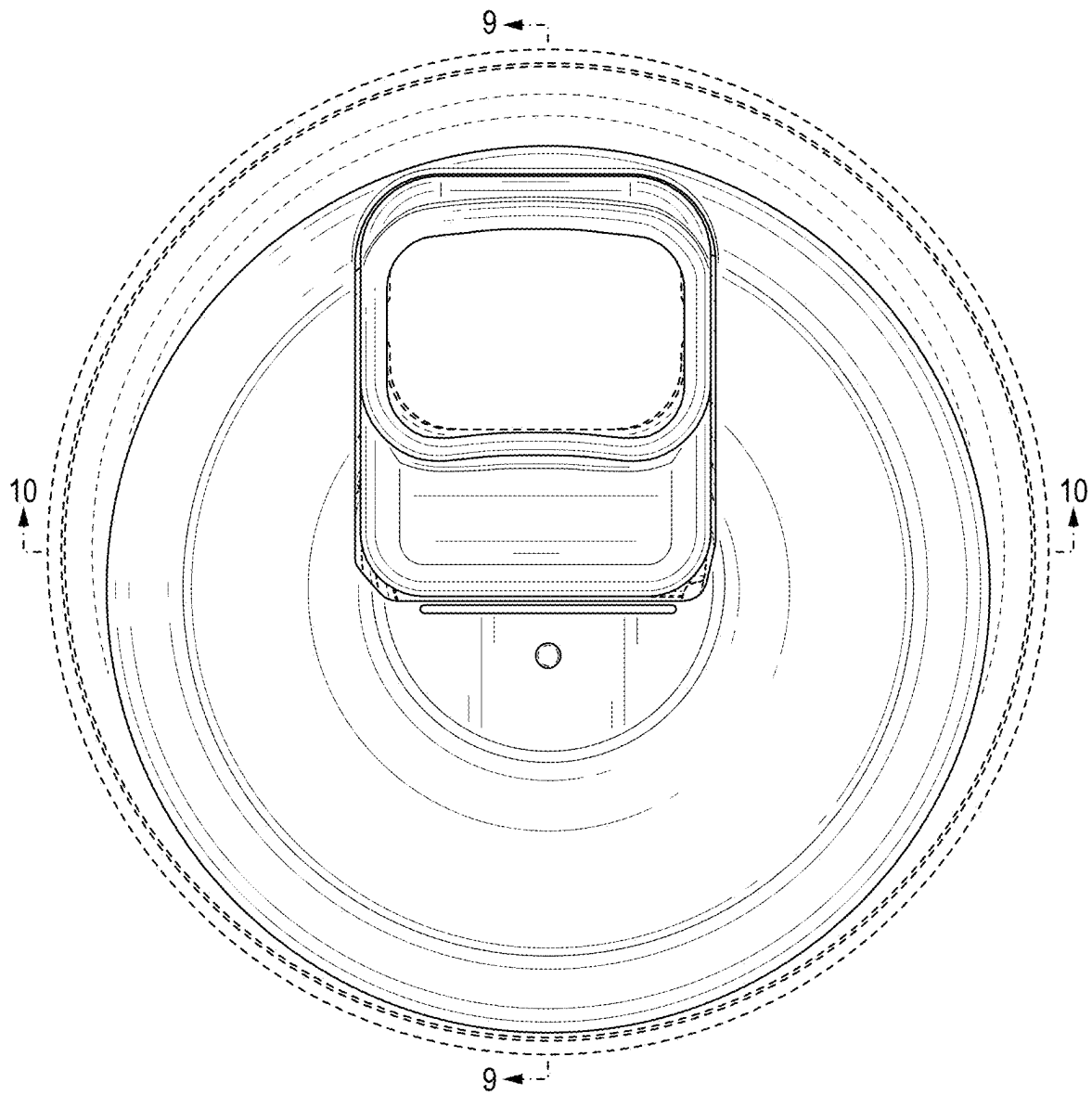


FIG. 5

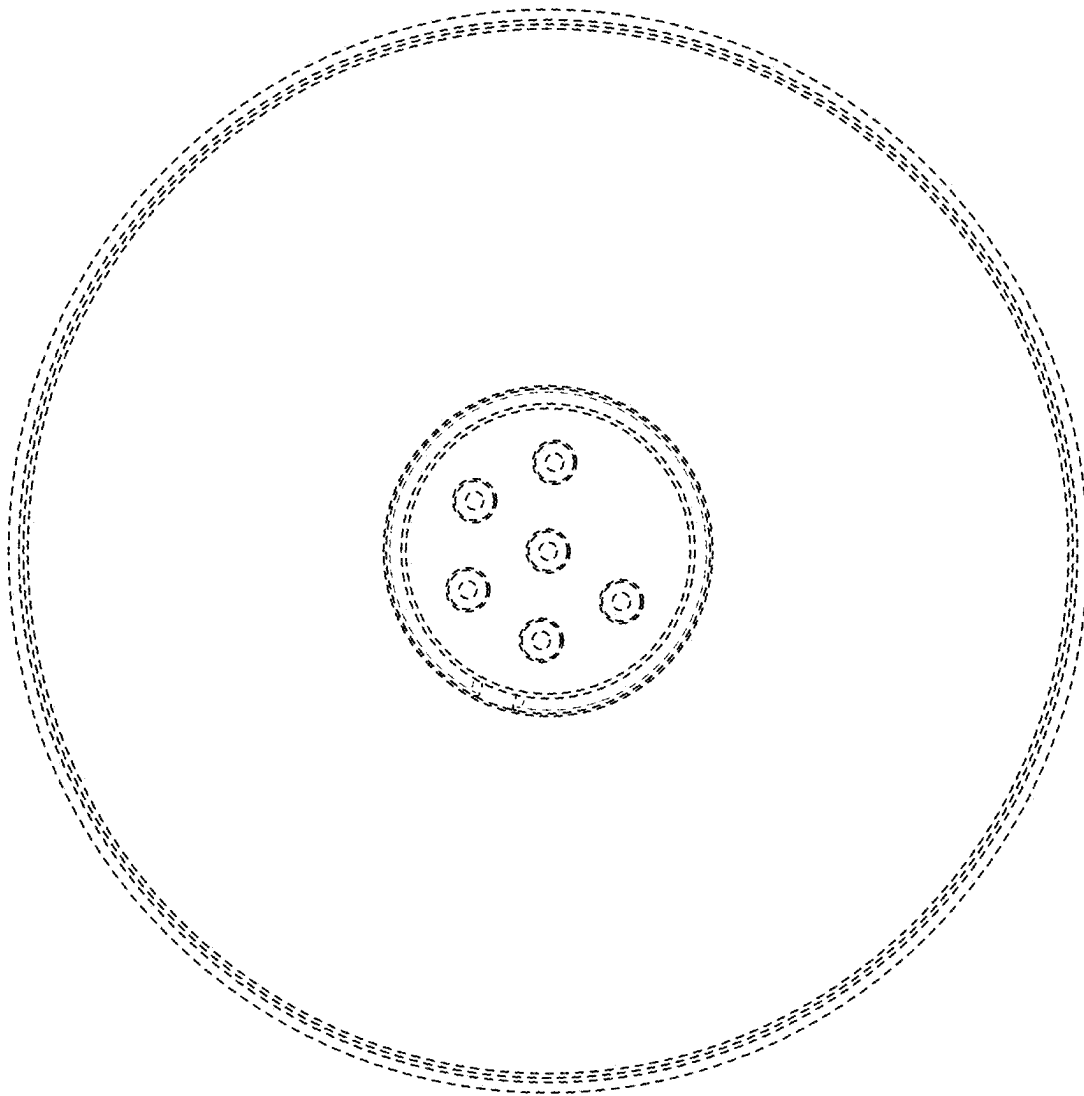


FIG. 6

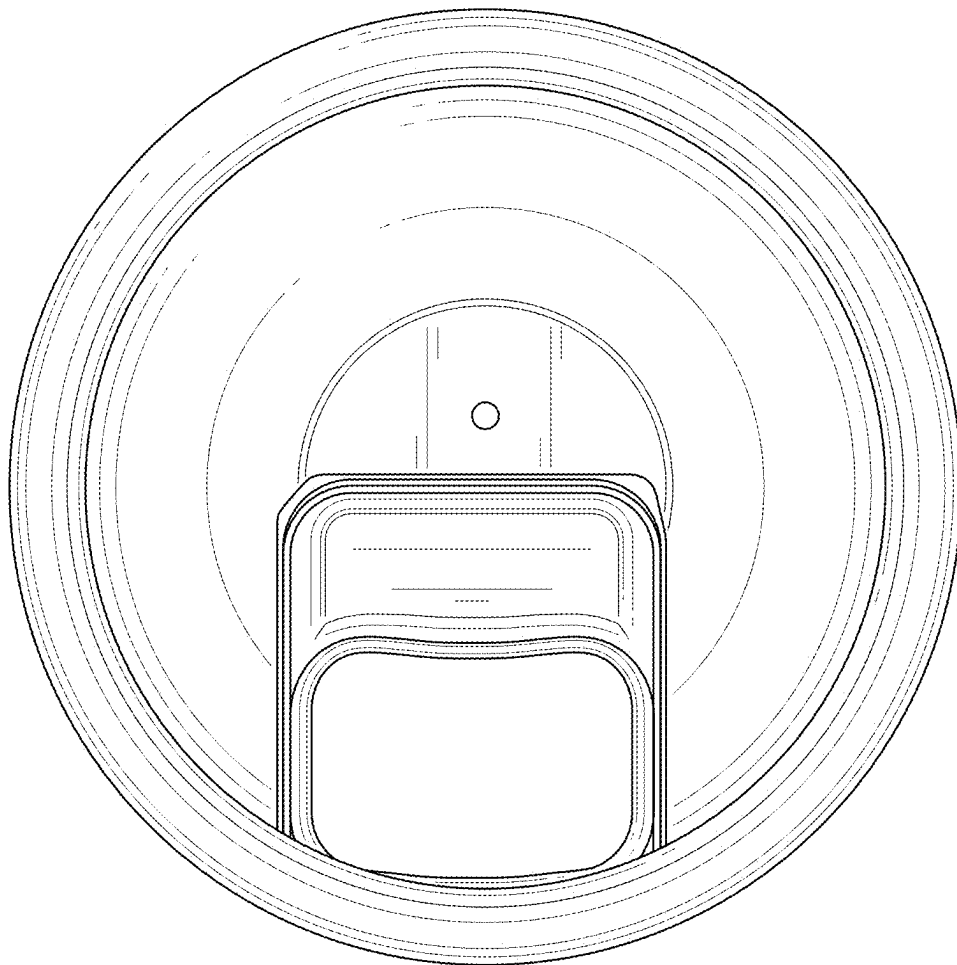


FIG. 7

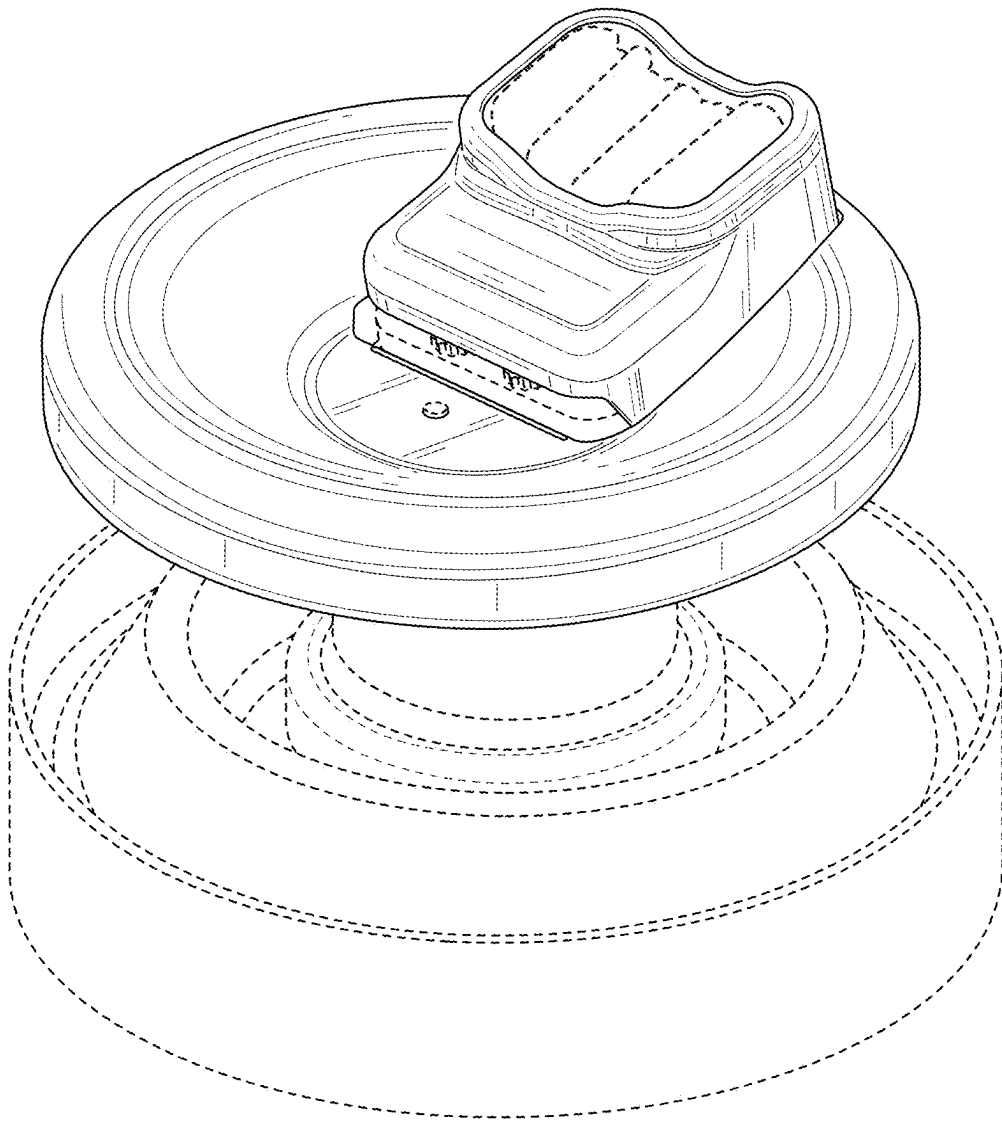


FIG. 8

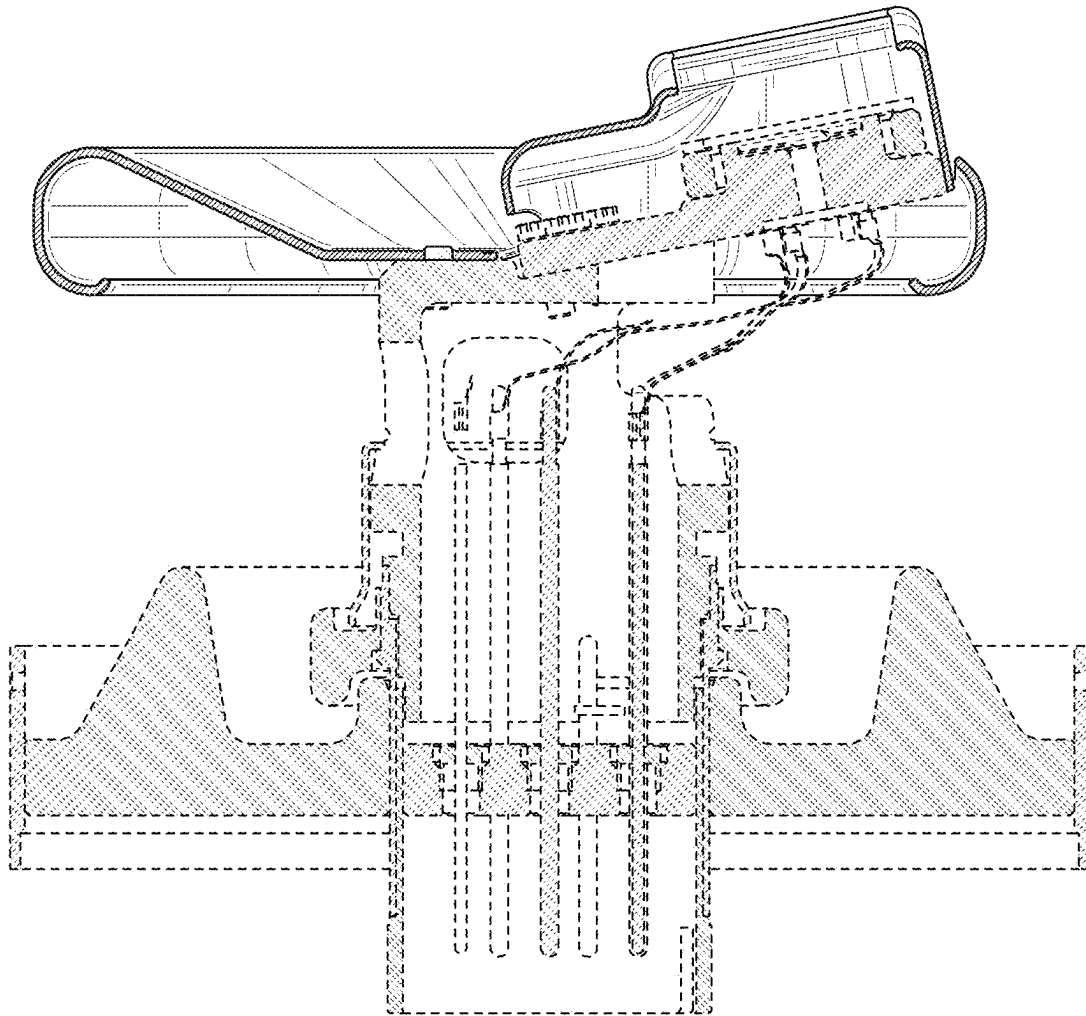


FIG. 9

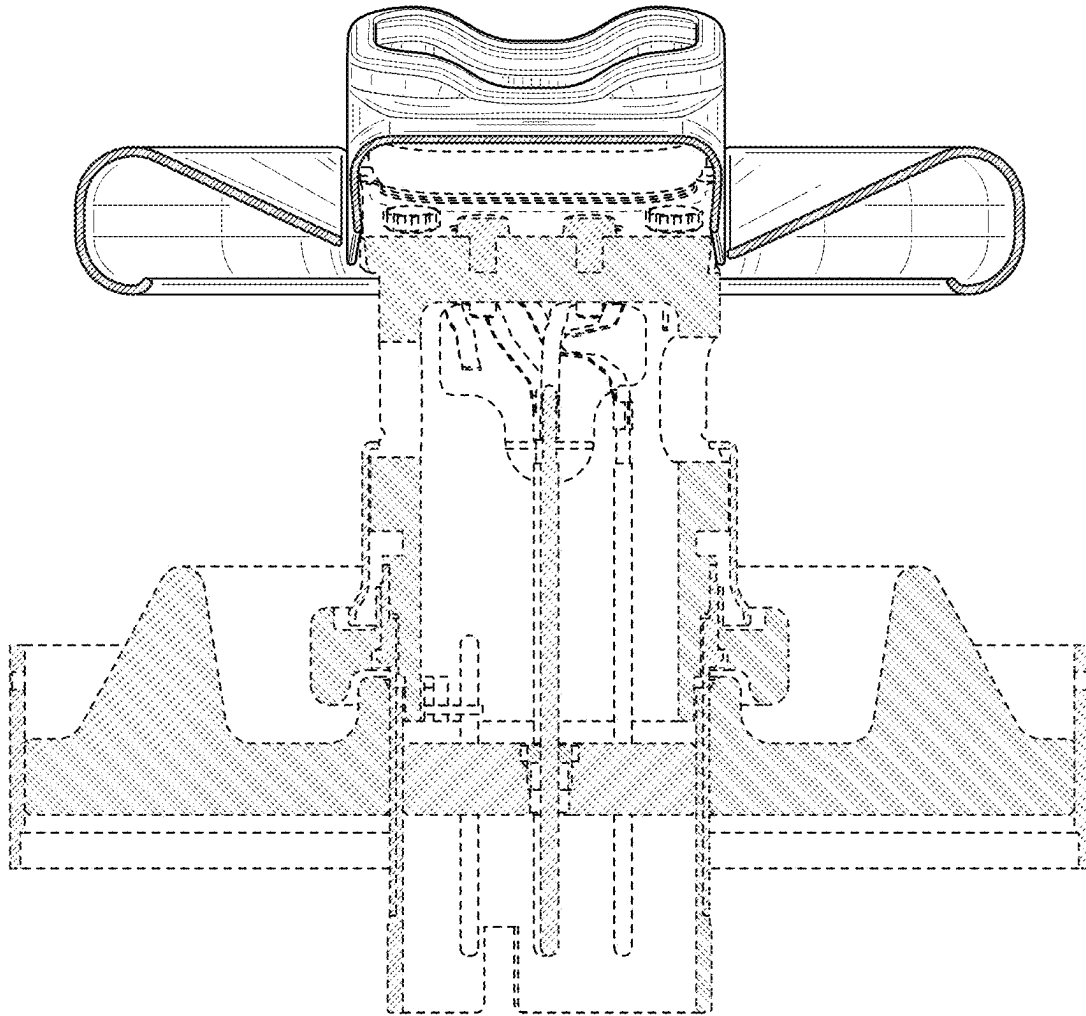


FIG. 10